Odd-sector, Pontryagin-space and index-law extensions of the Connes–van Suijlekom real-zeros theorem

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Abstract

Connes and van Suijlekom [1] proved that if the quadratic form attached to a real distribution on $[0, L]$ defines a lower-bounded self-adjoint operator whose lowest eigenvalue is simple, isolated and carries an *even* eigenfunction ξ, then the Fourier transform $\hat{\xi}$ has only real zeros. We resolve the two parity steps of their argument and obtain three extensions. **Theorem 1:** if a simple eigenvalue has κ eigenvalues below it, the transform of its eigenfunction has at most κ pairs of non-real zeros; for $\kappa = 0$ all zeros are real for *odd* as well as even ground states. Theorem 1 is the analogue, for the Connes–van Suijlekom class, of Gueguen's theorem on eigenpolynomials of Hermitian Toeplitz matrices, and its two-sided form reproduces Gueguen's count exactly. **Theorem 2:** the ground state of the odd sector alone has all its zeros on the real or the imaginary axis; positivity is required on the odd subspace only. **Theorem 3:** when exactly one eigenvalue lies below the odd ground state, the number of purely imaginary pairs (zero or one) is fixed by comparing the signs of the first moment and of the boundary behaviour. All results hold in finite dimension and, under the hypotheses of [1, Thm 6.1] together with a Rayleigh–Ritz lemma, at operator level. They are verified on random matrices of the class with negative controls and illustrated on truncated Weil quadratic forms.

Added in version 2.0. The zeros of the ground-state polynomial P_ξ interlace with those of the companion $P_{\beta\odot\xi}$ built from the distribution itself (Theorem 4). For every simple eigenvalue λ an exact index law holds (Theorem 5): the number of pairs of non-real zeros of $\hat{\xi}$ plus the number of real zeros of negative type—those where $P'_\xi P_{\beta\odot\xi} > 0$—equals the number $\kappa(\lambda)$ of eigenvalues below λ, with two explicit terms added in the odd sector. Equivalently, $Q - \lambda$ is congruent to minus the Bezout matrix of P_ξ and $P_{\beta\odot\xi}$ (Corollary 5.1). For truncated Weil forms at $q = 13, 29, 61$ the law holds in all 237 eigenvectors with interval-certified signs, and all eigenvectors with eigenvalue below 10^{-3} are certified to have only real, simple zeros (Tables 3–4).

Added in version 2.1. A resolvent version of the index law (Theorem 6) reads the level of the spectrum from zeros: for $x = (Q - \mu)^{-1} e_0$ the number of pairs of non-real zeros of P_x, plus its real zeros of negative type, plus $[x_0 < 0]$, equals the number of eigenvalues below μ. At $\mu = 0$ this is a zero-location criterion for $\lambda_0 > 0$. For the truncated Weil forms the criterion holds in all nine cells, and for matrices shifted to have 1, 2 or 4 negative eigenvalues it returns exactly these counts (Table 5).

Scope. Proved: Theorems 1–6 (finite dimension) and Theorems 1′–3′ (operator level). Computer-verified (interval arithmetic): zero certificates for truncated Weil forms (Tables 3–5). Measured: random-matrix verification (Table 1), Weil eigenvalue data (Table 2), excited-state observations (Section 15). Continuation: uniform isolation along windows (Section 1.1) and the capacity law of Section 15.

Keywords: Weil quadratic form, real zeros, Carathéodory–Fejér, Gueguen theorem, Pontryagin space, Loewner matrix, odd sector, Bezout matrix, interlacing, index law, interval arithmetic, resolvent, positivity criterion.

1 Introduction and results

The theorem of Connes and van Suijlekom (CvS) [1, Thm 6.1] states: if the quadratic form defined by a real distribution on $[0, L]$ yields a lower-bounded self-adjoint operator whose spectrum has a simple isolated minimum with *even* eigenfunction ξ, then all zeros of $\hat{\xi}$ are real. The finite-dimensional core of the proof [1, Thm 5.6] uses parity twice: $\langle \beta \,|\, \xi \rangle = 0$ for even ξ, and the normalisation $\langle \eta \,|\, \xi \rangle = 1$. For odd ξ the pairing $\langle \eta \,|\, \xi \rangle$ vanishes by parity, so the second step calls for a new ingredient.

This note supplies it. Every vector $\varphi \in \operatorname{span}\{\eta, \beta\}$ serves as auxiliary vector (Lemma 3); a finite negative index replaces positivity (Theorem 1); D^2 preserves the odd sector and keeps the argument inside it (Theorem 2); and the two results combined fix the purely imaginary zeros through a single sign condition (Theorem 3). Theorem 1 is the analogue for the CvS class of Gueguen's theorem [10,12] on Hermitian Toeplitz matrices; Theorems 2 and 3 are, to our knowledge, new.

1.1 Context

In the CvS programme the ground state of the truncated Weil form has a real-rooted transform [1] whose zeros approximate the low-lying zeros of ζ [3]. Uniform convergence of these transforms to Ξ on compact sets, as the window grows, would transfer real-rootedness to Ξ through Hurwitz's theorem. The present results extend the family of real-rooted approximants to the odd sector and give a counted control of excited states; the uniform version of the isolation hypothesis across windows is the next step of the programme.

1.2 Added in version 2.0

Theorem 4 shows that the ground-state polynomial interlaces with the companion $P_{\beta \odot \xi}$. Theorem 5 turns the inequality of Theorem 1 into an equality: the negative index $\kappa(\lambda)$ is carried exactly by non-real zero pairs and by real zeros of negative type, and the type of each real zero is read off from the sign of $P_\xi' P_{\beta \odot \xi}$ (Lemma 6). Section 14 uses both results as certificates for truncated Weil forms, and Section 15 records where the additional zeros of the excited states are located.

1.3 Added in version 2.1

The zeros of eigenvectors are blind to the level of the spectrum: $Q + cI$ has the same eigenvectors, polynomials and counts as Q. Theorem 6 supplies the missing level through the resolvent vector $(Q - \mu)^{-1} e_0$; at $\mu = 0$ it turns the inequality $\lambda_0 > 0$ into a statement about the location of zeros.

2 Related work

Toeplitz matrices. Gueguen [10] proved: if λ_r is the r-th eigenvalue of a Hermitian Toeplitz matrix T_n and lies outside the spectrum of T_{n-1}, then the associated eigenpolynomial has at least $|n - 2r + 1|$ zeros on the unit circle (see Trench [12, Thm 1]). Delsarte, Genin and Kamp [11] extended the count to multiple eigenvalues; for the extreme eigenvalues all zeros lie on the circle, a fact well known in signal processing [13]; see also the survey of Genin [14]. Theorem 1 and its two-sided form (Remark 5.2) carry this count to the CvS class: Loewner-type matrices in place of Toeplitz matrices, the real axis in place of the unit circle.

CvS class. The even case $\kappa = 0$ of Theorem 1 is [1, Thm 5.6/6.1]; the odd case $\kappa = 0$ extends to every N the explicit computation of [1, App. B.3] for $N = 2$. The tool is Pontryagin's theorem [2,6]; the rank-one method of [1] is carried to the family $\varphi \in \operatorname{span}\{\eta, \beta\}$ and to indefinite forms.

Indefinite weights. Zero counting through inertia for polynomials orthogonal with respect to indefinite weights goes back to Krein [15] and Krein–Langer [4]; see also Ellis–Gohberg [5]. Generalized Nevanlinna functions and Loewner theory in the class N_κ provide the general framework [7,8]. On the number-theoretic side, Zhu [9] certifies in compact windows the sign change of the pole term in the odd sector and odd-sector positivity, and observes the alternation $\lambda_1^{\text{even}} < \lambda_1^{\text{odd}} < \lambda_2^{\text{even}}$ at every scanned window [9, §6.3]; this is the regime $\kappa = 1$ of Theorem 3.

3 Definitions

Let $N \geq 1$, $I = \{-N, \dots, N\}$, and let e_j be the standard basis of $\mathbb{R}^I$. A real symmetric matrix Q is of *CvS type* [1, eq. (11)] if

$$q_{ii} = a_i, \qquad q_{ij} = \frac{b_i - b_j}{i - j} \ (i \neq j), \qquad a_{-i} = a_i, \quad b_{-i} = -b_i. \tag{1}$$

Further $\gamma e_j = e_{-j}$ (the reflection $x \mapsto L - x$), $De_j = j\, e_j$ (the generator of translations), $\eta = \sum_j e_j$, $\beta = \sum_j b_j e_j$; thus $\gamma\eta = \eta$ and $\gamma\beta = -\beta$. Write $X \wedge Y := |X\rangle\langle Y| - |Y\rangle\langle X|$. For $\xi \in \mathbb{R}^I$ define the root polynomial

$$P_\xi(s) = \sum_{k \in I} \xi_k \prod_{j \neq k} (j - s). \tag{2}$$

Let $\xi(x) = \sum_k \xi_k e^{2\pi i k x}$ on $[0, 1]$, extended by 0. Its Fourier transform is [1, Thm 5.6(ii)]

$$\hat{\xi}(z) = 2e^{-iz/2} \sin \tfrac{z}{2} \sum_k \frac{\xi_k}{z - 2\pi k} = -\frac{1}{\pi}\, e^{-iz/2} \sin \tfrac{z}{2}\, \frac{P_\xi(z/2\pi)}{\prod_j (j - z/2\pi)}. \tag{3}$$

Since the zeros of $\sin(z/2)$ are real, the non-real zeros of $\hat{\xi}$ are exactly the non-real roots of $P_\xi(z/2\pi)$. Finally $\kappa(\lambda)$ denotes the number of eigenvalues of Q strictly below λ (with multiplicity); $Q_\lambda := Q - \lambda I$ is again of CvS type.

4 Lemmas

Lemma 1 (CvS Lemma 5.1)**.** *$Q\gamma = \gamma Q$, $D\gamma = -\gamma D$ and*

$$[D, Q] = DQ - QD = \beta \wedge \eta. \tag{4}$$

The same identity holds for Q_λ.

Lemma 2 (Rank-one determinant)**.** *Let $c := \langle \varphi \,|\, \xi \rangle \neq 0$ and $D'_\varphi := D - |D\xi\rangle\langle\varphi|/c$. Then $D'_\varphi \xi = 0$ and*

$$\det(D'_\varphi - s) = -\frac{s}{c}\, P_{\varphi \odot \xi}(s), \qquad (\varphi \odot \xi)_k = \varphi_k \xi_k. \tag{5}$$

Proof. By the matrix determinant lemma, $\det(D'_\varphi - s) = \det(D - s)\big(1 - \langle\varphi|(D - s)^{-1}|D\xi\rangle/c\big)$. Since $(D - s)^{-1} D = I + s(D - s)^{-1}$, the bracket equals $-s\langle\varphi|(D - s)^{-1}|\xi\rangle/c$; multiplying by $\det(D - s) = \prod_j (j - s)$ gives (5). □

Lemma 3 (Q_λ-symmetry criterion)**.** *Let $Q_\lambda \xi = 0$. For every $\varphi \in \operatorname{span}\{\eta, \beta\}$ with $c = \langle \varphi \,|\, \xi \rangle \neq 0$ the matrix $Q_\lambda D'_\varphi$ is symmetric, i.e. D'_φ is self-adjoint for the form of Q_λ.*

Proof. Symmetry is equivalent to $Q_\lambda D - D Q_\lambda = (v \wedge \varphi)/c$ with $v := Q_\lambda D\xi$. Applying (4) to ξ gives $v = \langle \beta \,|\, \xi \rangle \eta - \langle \eta \,|\, \xi \rangle \beta$, while the left side equals $-[D, Q_\lambda] = \eta \wedge \beta$. For $\varphi = x\eta + y\beta$,

$$v \wedge \varphi = \big(x\langle \eta \,|\, \xi \rangle + y\langle \beta \,|\, \xi \rangle\big)\, \eta \wedge \beta = c\, \eta \wedge \beta, \tag{6}$$

so the condition holds for all x, y. (CvS use $\varphi = \eta$, $c = 1$.) □

Lemma 4 (Non-degeneracy)**.** *Let λ be simple with eigenvector $\xi \notin \mathbb{R}e_0$. Then ξ has definite parity, and: for even ξ, $\langle \beta \,|\, \xi \rangle = 0$ and $\langle \eta \,|\, \xi \rangle \neq 0$; for odd ξ, $\langle \eta \,|\, \xi \rangle = 0$ and $\langle \beta \,|\, \xi \rangle \neq 0$.*

Proof. Q commutes with γ and λ is simple, so $\gamma\xi = \pm\xi$. The vanishing pairing comes from parity. If the other pairing vanished as well, then $v = 0$, i.e. $D\xi \in \ker Q_\lambda = \mathbb{R}\xi$. Since $D\xi$ has the opposite parity, $D\xi = 0$, hence $\xi \in \ker D = \mathbb{R}e_0$, a contradiction. □

5 Theorem 1: Pontryagin–CvS

Theorem 1. *Let Q be of CvS type, λ a simple eigenvalue with eigenvector ξ, and $\kappa = \kappa(\lambda)$. Then P_ξ has at most κ pairs of non-real roots (counted with multiplicity); hence $\hat{\xi}$ has at most κ pairs of non-real zeros.*

Corollary 1.1 ($\kappa = 0$)**.** *If the lowest eigenvalue is simple, all zeros of $\hat{\xi}$ are real, for odd as well as even eigenvectors.*

Proof of Theorem 1. (1) If $\xi \in \mathbb{R}e_0$, then $\hat{\xi} \propto e^{-iz/2}\sin(z/2)/z$ and all zeros are real. Assume $\xi \notin \mathbb{R}e_0$.

(2) Put $\varphi_t := \eta + t\beta$ and $c_t := \langle \varphi_t \,|\, \xi \rangle = \langle \eta \,|\, \xi \rangle + t\langle \beta \,|\, \xi \rangle$. By Lemma 4, for even ξ, $c_t = \langle \eta \,|\, \xi \rangle \neq 0$ for every t; for odd ξ, $c_t = t\langle \beta \,|\, \xi \rangle \neq 0$ for every $t \neq 0$.

(3) Let $D_t := D'_{\varphi_t}$. By Lemma 2, $D_t\xi = 0$; by Lemma 3, D_t is self-adjoint for the form of Q_λ.

(4) On $E := \mathbb{R}^I/\mathbb{R}\xi$ the form $[x, y] := \langle Q_\lambda x \,|\, y \rangle$ is non-degenerate ($\ker Q_\lambda = \mathbb{R}\xi$) with exactly κ negative squares. Since D_t annihilates $\mathbb{R}\xi$, it descends to a self-adjoint operator $\bar{D}_t$ on E.

(5) Pontryagin's theorem in finite dimension [2,6]: a self-adjoint operator on a space with κ negative squares has eigenvalues in the open upper half-plane of total algebraic multiplicity at most κ.

(6) In a basis consisting of ξ and a complement, D_t is block triangular: $\det(D_t - s) = -s\det(\bar{D}_t - s)$. By (5), $\det(\bar{D}_t - s) = P_{\varphi_t \odot \xi}(s)/c_t$, so $P_{\varphi_t \odot \xi}$ has at most κ pairs of non-real roots.

(7) $P_{\varphi_t \odot \xi} = P_\xi + t\,P_{\beta \odot \xi}$ converges coefficientwise to P_ξ as $t \to 0$, and $P_\xi(k) = \xi_k \prod_{j \neq k}(j - k)$ shows $P_\xi \not\equiv 0$. By Hurwitz's theorem, P_ξ has at most κ pairs of non-real roots. (For even ξ one takes $t = 0$ directly.)

(8) By (3), the non-real zeros of $\hat{\xi}$ are the non-real roots of $P_\xi(z/2\pi)$. □

Remark 5.1. For $\kappa = 0$ the space E is Euclidean and the proof reduces to that of CvS. In the odd case $\langle \eta \,|\, \xi \rangle = 0$ by parity; the family φ_t and the limit $t \to 0$ take the place of the normalisation $\langle \eta \,|\, \xi \rangle = 1$.

Remark 5.2 (Two-sided form). $-Q$ is again of CvS type ($a \mapsto -a$, $b \mapsto -b$) with the same eigenvector, so Theorem 1 also bounds the count by the number of eigenvalues above λ. If λ is the r-th eigenvalue and $n = 2N + 1$, then P_ξ has at most $\min(r - 1, n - r)$ pairs of non-real roots: exactly Gueguen's count for Toeplitz matrices [10,12]. For even ξ, where $\deg P_\xi = n - 1$, this means at least $|n - 2r + 1|$ real roots; for odd ξ the leading coefficient $\langle \eta \,|\, \xi \rangle$ vanishes, $\deg P_\xi \leq n - 2$, and the bound reads $|n - 2r + 1| - 1$ (the remaining root sits at infinity).

6 Theorem 2: the odd sector

Theorem 2. *Let Q be of CvS type and let λ_o be the minimum of Q on the odd subspace H_o, simple within H_o, with eigenvector $\xi \in H_o$. Then every zero of $\hat{\xi}$ is real or purely imaginary. The proof uses positivity on H_o only.*

The key observation: D maps the odd sector to the even sector, while D^2 maps the odd sector to itself; the argument therefore stays inside the odd sector.

Proof. (1) H_o has the basis $o_j := (e_j - e_{-j})/\sqrt{2}$, $j = 1, \dots, N$. Q preserves H_o; so does $A := D^2$, with $Ao_j = j^2 o_j$, a simple spectrum on H_o.

(2) By Lemma 1, $[D^2, Q] = D[D, Q] + [D, Q]D = |D\beta\rangle\langle\eta| - |D\eta\rangle\langle\beta| + |\beta\rangle\langle D\eta| - |\eta\rangle\langle D\beta|$. Here η and $D\beta$ are even, while β and $\tilde{\eta} := D\eta = \sum_j j\, e_j$ are odd. Compressing to H_o removes the even terms:

$$[A, Q_o] = \beta \wedge \tilde{\eta}, \tag{7}$$

which is Lemma 1 with (D, η) replaced by $(A, \tilde{\eta})$.

(3) $Q_{o,\lambda} := Q_o - \lambda_o$ is positive semidefinite on H_o with kernel $\mathbb{R}\xi$. Lemmas 2–3 carry over with $\varphi_t := \tilde{\eta} + t\beta$ and $c_t := \langle \varphi_t \,|\, \xi \rangle$. If both pairings vanish, then $A\xi \in \mathbb{R}\xi$, so $\xi = \pm o_j$; then $\hat{\xi} \propto e^{-iz/2} \sin(z/2)/(z^2 - 4\pi^2 j^2)$, whose zeros are real.

(4) $H_o/\mathbb{R}\xi$ is Euclidean, so the descended operator has real spectrum. With $\tilde{\xi}_k := \langle o_k \,|\, \xi \rangle$, the polynomial

$$R_t(\sigma) = \sum_{k=1}^{N} \langle o_k \,|\, \varphi_t \rangle\, \tilde{\xi}_k \prod_{j \neq k} (j^2 - \sigma) \tag{8}$$

is real-rooted for small $t \neq 0$. By $t \to 0$ and Hurwitz's theorem R_0 is real-rooted; since $\langle o_k \,|\, \tilde{\eta} \rangle = \sqrt{2}\, k$, $R_0(\sigma) \propto \sum_k k\, \tilde{\xi}_k \prod_{j \neq k} (j^2 - \sigma)$.

(5) For odd ξ we have $\xi_{-k} = -\xi_k$ and

$$\sum_n \frac{\xi_n}{z - 2\pi n} = \sum_{k \geq 1} \xi_k \, \frac{4\pi k}{z^2 - 4\pi^2 k^2} = -\frac{1}{\pi} \sum_k \frac{k\, \xi_k}{k^2 - \sigma}, \qquad \sigma := \frac{z^2}{4\pi^2}. \tag{9}$$

Hence for every zero of $\hat{\xi}$ off the zeros of $\sin(z/2)$, $\sigma = z^2/4\pi^2$ is a real root of R_0: z is real for $\sigma \geq 0$ and purely imaginary for $\sigma < 0$. □

7 Theorem 3: the dichotomy at $\kappa = 1$

Theorem 3. *Under the hypotheses of Theorem 2, let n_- be the number of negative roots of R_0, i.e. the number of pairs of purely imaginary zeros of $\hat{\xi}$. With the Fourier coefficients of positive index put*

$$\mu_1 := \sum_{k=1}^{N} \frac{\xi_k}{k}, \qquad \mu_2 := \sum_{k=1}^{N} k\, \xi_k, \qquad \mu_1 \mu_2 \neq 0. \tag{10}$$

(i) n_- is even $\iff$ $\mu_1 \mu_2 > 0$.

(ii) If moreover λ_o is a simple eigenvalue of Q on the full space with $\kappa(\lambda_o) = 1$, then all zeros of $\hat{\xi}$ are real $\iff$ $\mu_1 \mu_2 > 0$; otherwise there is exactly one purely imaginary pair $\pm i y_0$.

Proof. R_0 has degree $N - 1$ and leading coefficient proportional to $(-1)^{N-1} \mu_2$ (positive factor); hence its sign as $\sigma \to -\infty$ is that of μ_2. Also $R_0(0) \propto \big(\prod_j j^2\big) \mu_1$. All roots are real (Theorem 2), so the parity of the number of roots in $(-\infty, 0)$ records whether these two signs differ; this is (i). For $\kappa = 1$, Theorem 1 bounds the number of non-real pairs by one, so $n_- \in \{0, 1\}$ and the parity determines it; this is (ii). □

Continuous form. On the window $[-h, h]$, for real odd ξ, the values on the imaginary axis give the real function

$$g(y) = \hat{\xi}(iy) = \int_{-h}^{h} \xi(u)\, e^{yu}\, du = \int_{-h}^{h} \xi(u) \sinh(yu)\, du, \qquad g(y) \approx y \int u\, \xi(u)\, du \quad (y \to 0^+). \tag{11}$$

For large y the sign of g is the sign of ξ near the endpoint h (Theorem 3′). By the intermediate value theorem, the parity of the number of purely imaginary pairs records whether the first moment $\int u\xi$ and the boundary sign differ; for $\kappa = 1$ this becomes an exact dichotomy. In a Legendre basis the first moment has the sign of x_0, and the boundary value $\xi(h)$ is a positive multiple of $\sum_n \sqrt{4n+3}\, x_n$.

8 Operator level

We work under the hypotheses of CvS Thm 6.1: $\mathcal{D}$ is a real distribution on $[0, L]$; the form defined by CvS (6) yields a lower-bounded self-adjoint operator A, and the trigonometric polynomials $\mathcal{T}$ form a core. E_N denotes the trigonometric polynomials with indices $|n| \le N$; the compression Q_N of the form to E_N is of CvS type [1, Props. 4.1–4.2]. $\mu_j(\cdot)$ denotes min-max values [16, §XIII.1] and q the form.

Lemma 5 (Rayleigh–Ritz below a gap)**.** *Assume that for some $\delta > 0$ the spectrum of A in $(-\infty, \lambda + \delta)$ consists of eigenvalues $\mu_1 \le \cdots \le \mu_k < \lambda$ and the simple eigenvalue $\lambda = \mu_{k+1}$. Then for large N: (i) $\mu_j(Q_N) \to \mu_j$ for $j \le k+1$, and $\mu_{k+2}(Q_N) \ge \lambda + \delta$; (ii) $\lambda_N := \mu_{k+1}(Q_N)$ is simple with exactly k eigenvalues below it; (iii) the normalised eigenvector $\xi_N \to \xi$ in L^2 (after a choice of sign), and ξ_N has the parity of ξ.*

Proof. (i) Since $E_N \subset E_{N+1}$, the min-max principle gives $\mu_j(Q_N) \ge \mu_j(Q_{N+1}) \ge \mu_j(A)$; moreover $\mu_{k+2}(A) \ge \lambda + \delta$. For the upper bound take orthonormal eigenvectors $\psi_1, \dots, \psi_{k+1}$ ($\psi_{k+1} = \xi$) and, by the core property, $\varphi_i \in \mathcal{T}$ close to them in graph norm. Then $G = (\langle \varphi_i \,|\, \varphi_j \rangle) \to I$ and $F = (q(\varphi_i, \varphi_j)) \to \operatorname{diag}(\mu_i)$; once all $\varphi_i \in E_N$, $\mu_j(Q_N)$ is bounded by the j-th generalised eigenvalue of the pair (F, G). (ii) follows from (i). (iii) Let Π_N be the spectral projection of Q_N onto its lowest $k+1$ eigenvalues, S the span of the φ_i, and P the projection onto the first $k+1$ eigenvectors of A. The Ky Fan inequality with gap reads

$$\operatorname{tr}\big(q|_S\big) \;\ge\; \sum_{j \le k+1} \mu_j(Q_N) + \big(\mu_{k+2}(Q_N) - \mu_{k+1}(Q_N)\big)\, \big\|(1 - \Pi_N) P_S\big\|_{HS}^2. \tag{12}$$

(Proof: with eigenvectors e_j of Q_N, $w_j = \|P_S e_j\|^2 \in [0,1]$, $\sum_j w_j = k+1$ and $t := \sum_{j>k+1} w_j$, one has $\sum_j \mu_j w_j \ge \sum_{j \le k+1} \mu_j - \mu_{k+1} t + \mu_{k+2} t$.) The left side tends to $\sum_{j \le k+1} \mu_j(A)$, the sum on the right is at least $\sum_{j \le k+1} \mu_j(A)$, and for large N the gap is at least $\delta/2$. Hence $\|(1 - \Pi_N) P_S\| \to 0$; as $P_S \to P$ and the ranks agree, $\|\Pi_N - P\| \to 0$. The same argument for the lowest k eigenvalues (gap $\lambda - \mu_k$) gives $\|\Pi_N^{(k)} - P^{(k)}\| \to 0$ for the corresponding projections. Since $\xi_N \in \operatorname{Ran} \Pi_N$ and $\xi_N \perp \operatorname{Ran} \Pi_N^{(k)}$, we get $\langle \xi_N \,|\, \psi_i \rangle \to 0$ for $i \le k$ and $|\langle \xi_N \,|\, \xi \rangle| \to 1$. Parity: Q_N commutes with γ, the eigenvector of a simple eigenvalue has definite parity, and convergence to ξ fixes it. □

Theorem 1′. *Under the hypotheses of Lemma 5, with $\kappa := k$, $\hat{\xi}$ has at most κ pairs of non-real zeros (counted with multiplicity). For $\kappa = 0$ all zeros are real, for odd as well as even ξ.*

Proof. For large N, Lemma 5 makes Theorem 1 applicable to ξ_N with κ. The Cauchy–Schwarz estimate $|\hat{\xi}_N(z) - \hat{\xi}(z)| \le \|\xi_N - \xi\|_2 \big(\int_0^L e^{2\operatorname{Im}(z)x} dx\big)^{1/2}$ (cf. [1, proof of Thm 3.1]) gives uniform convergence on compact sets. If $\hat{\xi}$ had $\kappa + 1$ pairs of non-real zeros, Hurwitz's theorem applied in small discs away from the real axis would give $\hat{\xi}_N$ as many for large N, a contradiction. □

Theorem 2′. *If the bottom λ_o of the restriction of A to odd functions is a simple isolated eigenvalue of that restriction, then all zeros of the transform $\hat{\xi}$ of its eigenfunction lie in $\mathbb{R} \cup i\mathbb{R}$.*

Proof. A commutes with γ, and the odd trigonometric polynomials $\frac{1}{2}(1-\gamma)\mathcal{T}$ form a core of the restriction. Lemma 5 with $k = 0$ applies in the odd sector, and Theorem 2 places the zeros of $\hat{\xi}_N$ in $\mathbb{R} \cup i\mathbb{R}$. By Hurwitz's theorem each zero of $\hat{\xi}$ is a limit of zeros of $\hat{\xi}_N$, and $\mathbb{R} \cup i\mathbb{R}$ is closed. □

Theorem 3′. *In addition to the hypotheses of Theorem 2′, let λ_o be a simple isolated eigenvalue of A on the full space, the spectrum of A below λ_o consisting of a single simple eigenvalue. Shift the window to $[-h, h]$ and assume: **(B)** ξ has constant sign almost everywhere on some $(h-\varepsilon, h)$, and h belongs to the essential support of ξ; **(M)** $m_1 := \int u\,\xi(u)\,du \neq 0$. Then all zeros of $\hat{\xi}$ are real $\iff$ m_1 and the boundary sign of ξ agree. Otherwise there is exactly one purely imaginary pair $\pm iy_0$, and it is simple.*

Proof. For odd ξ, $g(y) := \hat{\xi}(iy) = \int \xi(u)e^{yu}\,du$ is real and odd. By Theorem 2′ the pairs $\pm iy$ of non-real zeros correspond exactly to the zeros y of g in $(0, \infty)$, and by Theorem 1′ with $\kappa = 1$ there is at most one such y, counted with multiplicity. Since $g'(0) = m_1$, the sign of g for small $y > 0$ is that of m_1. For large y, with a suitable $\varepsilon' < \varepsilon$, the term $\int_{h-\varepsilon'}^{h} \xi e^{yu}$ exceeds $c\,e^{y(h-\varepsilon')}$ in absolute value ($c > 0$) while the remainder is $O(e^{y(h-\varepsilon)})$; the sign is the boundary sign of ξ. The zeros of the analytic function g are isolated; the parity of their number in $(0, \infty)$ records whether the two end signs differ, and since there is at most one, it is determined. □

9 Numerical verification

Setup: $N = 5$ (11×11 matrices), mpmath with 60 digits, eigenvectors by `eigsy`, roots by `polyroots`. The parameters a, b are drawn from $\mathcal{N}(0, 1)$, together with a diagonally dominant variant ($a \times 3$, $b \times 0.3$). Negative control: random γ-symmetric matrices outside the CvS class. Results in Table 1.

Table 1: Random-matrix verification of Theorems 1–3 (17 October 2026).

Claim	CvS class	Negative control
T1: non-real pairs $> j$ for the j-th eigenvector	0 / 1320	102 / 660
T1 bound attained (pairs $= j$, $j \geq 1$)	189 / 1200	–
T1 two-sided: pairs $> \min(j, n-1-j)$	0 / 660	196 / 660
Two-sided bound attained (upper half)	183 / 300	–
T1, $\kappa = 0$: odd ground states real-rooted	29 / 29	4 / 18
T2: zeros off $\mathbb{R} \cup i\mathbb{R}$	0 / 120	57 / 120
T3: dichotomy violations ($\kappa = 1$ cases)	0 / 84	–
T3: cases with a purely imaginary pair	44 / 120	–

The last row shows that purely imaginary zeros occur frequently in the CvS class: the second alternative of Theorem 2 is a genuine one, and real-rootedness statements are therefore issued together with an imaginary-axis scan.

10 Illustration: truncated Weil forms

The truncated Weil form fits the CvS framework: the explicit-formula distribution consists of archimedean, prime and pole terms, the latter with kernel $2\cosh(x/2)$. Our Legendre-basis computations are Rayleigh–Ritz approximations of the same operator. In the three windows measured, the odd ground state is the second eigenvalue of the full operator (Table 2).

Table 2: Eigenvalue ordering of truncated Weil forms in Legendre bases ($m = 80$ per sector, quadrature $F = 400$, working precision 140 digits); all values lie above the precision floor $\approx 1.4 \cdot 10^{-116}$.

q	even ground	odd ground	second even	κ(odd ground)
13	$1.1299 \cdot 10^{-57}$	$6.3232 \cdot 10^{-54}$	$2.1100 \cdot 10^{-50}$	1
29	$5.3617 \cdot 10^{-86}$	$1.5911 \cdot 10^{-82}$	$5.6087 \cdot 10^{-79}$	1
61	$7.6248 \cdot 10^{-103}$	$1.5315 \cdot 10^{-100}$	$5.4350 \cdot 10^{-97}$	1

- Sign condition ($m = 60$, $F = 260$, 100 digits): for the odd ground state at $q = 13$ the first moment and the boundary sign are both positive; at $q = 29$ both are negative. Theorem 3′, with (B) and (M) as measured, predicts that all zeros are real.
- Imaginary-axis scan ($m = 60$), $y \in [10^{-3}, 316]$, for the ground states of both sectors at $q = 13$ and 29: sign-constant.
- Argument-principle count ($m = 60$) in $[2, b] \times [-1.5,\ 1.5]$: odd ground state 22/22 ($q = 13$) and 18/18 ($q = 29$) real roots.

Simplicity, isolation and $\kappa = 1$ are measured in the Legendre truncations above; the sign condition and the imaginary-axis scan at $q = 61$ are the subject of the next run.

11 Theorem 4: interlacing at the ground state

Theorem 4. *Let Q be of CvS type and let its lowest eigenvalue λ be simple, with eigenvector $\xi \notin \mathbb{R}e_0$. Then P_ξ and $P_{\beta\odot\xi}$ have only real zeros, and these zeros interlace. For odd ξ the zero at $s = 0$, common to both polynomials by parity, is removed once from each before comparing.*

Proof. In the proof of Theorem 1 with $\kappa = 0$, every member of the pencil $P_{\varphi_t\odot\xi} = P_\xi + t\,P_{\beta\odot\xi}$ is real-rooted, for all real t (even ξ) or all $t \neq 0$ (odd ξ). Hurwitz's theorem at $t \to 0$ and $t \to \infty$ extends this to the whole real pencil, including P_ξ and $P_{\beta\odot\xi}$. By the Hermite–Kakeya–Obreschkoff theorem [17], two real polynomials whose real pencil is entirely real-rooted have interlacing zeros. For even ξ the degrees are $2N$ and $2N - 1$; for odd ξ, P_ξ has a simple and $P_{\beta\odot\xi}$ a double zero at $s = 0$, and the statement refers to the quotients by the common factor s. □

A second proof, with an explicit sign, is the case $\kappa = 0$ of Theorem 5: every real zero s of P_ξ then satisfies $P'_\xi(s)P_{\beta\odot\xi}(s) < 0$, so $P_{\beta\odot\xi}$ changes sign between consecutive zeros of P_ξ.

12 Theorem 5: the index law

For real $s \notin I$ put $\Delta(s) = \prod_{j\in I}(j - s)$, $w_s = (D - s)^{-1}\xi$, $v_s = (D - s)^{-1}D\xi = \xi + s\,w_s$ and

$$F(s) = \langle \eta \,|\, w_s \rangle = \frac{P_\xi(s)}{\Delta(s)}, \qquad G(s) = \langle \beta \,|\, w_s \rangle = \frac{P_{\beta\odot\xi}(s)}{\Delta(s)}. \tag{13}$$

Lemma 6 (Norm identity)**.** *Let $Q_\lambda\xi = 0$. For every real $s \notin I$,*

$$Q_\lambda(v_s, v_s) = s^2\big(F(s)G'(s) - F'(s)G(s)\big). \tag{14}$$

At a zero s of P_ξ this reads $Q_\lambda(v_s, v_s) = -s^2\,P'_\xi(s)\,P_{\beta\odot\xi}(s)/\Delta(s)^2$.

Proof. Since $Q_\lambda\xi = 0$, $Q_\lambda(v_s, v_s) = s^2 Q_\lambda(w_s, w_s)$. From $(D - s)w_s = \xi$ follows $Q_\lambda D w_s = sQ_\lambda w_s$, and (4) gives $Q_\lambda D - DQ_\lambda = \eta \wedge \beta$. Hence $(D - s)Q_\lambda w_s = F(s)\,\beta - G(s)\,\eta$. Pairing $Q_\lambda w_s = (D - s)^{-1}(F\beta - G\eta)$ with w_s and using $(D - s)^{-1}w_s = (D - s)^{-2}\xi = \frac{d}{ds}w_s$ yields $Q_\lambda(w_s, w_s) = FG' - F'G$. □

Lemma 7 (Pontryagin balance)**.** *Let V be a finite-dimensional real space with a non-degenerate symmetric form of negative index κ, and let T be self-adjoint for this form with simple eigenvalues only. Then κ equals the number of pairs of non-real eigenvalues plus the number of real eigenvalues whose eigenvectors have negative norm.*

Proof. Root subspaces of eigenvalues z, w with $z \neq \bar{w}$ are orthogonal, so V is the orthogonal sum of the eigenlines of the real eigenvalues and of the real two-dimensional invariant planes of the non-real pairs. Each such plane is spanned by the real and imaginary parts of an isotropic eigenvector and has signature $(1,1)$; each real eigenline is non-degenerate. Adding negative indices gives the claim [6, 19]. □

Theorem 5 (Index law)**.** *Let Q be of CvS type, λ a simple eigenvalue with eigenvector $\xi \notin \mathbb{R}e_0$, and $\kappa = \kappa(\lambda)$. Let N_c be the number of pairs of non-real zeros of P_ξ and*

$$d(\xi) = \#\big\{ s \in \mathbb{R} \setminus \{0\} : \; P_\xi(s) = 0, \; P'_\xi(s)\, P_{\beta\odot\xi}(s) > 0 \big\}. \tag{15}$$

(a) If ξ is even and the zeros of P_ξ are simple and lie outside I, then

$$N_c + d(\xi) = \kappa. \tag{16}$$

(b) If ξ is odd, the non-zero zeros of P_ξ are simple and lie outside I, and the pencil operators of the proof have simple spectrum for small $t \neq 0$, then

$$N_c + d(\xi) + \varepsilon_0 + \varepsilon_\infty = \kappa, \tag{17}$$

where $\varepsilon_0 = 1$ if $\langle e_0 \,|\, (Q^+ - \lambda)^{-1} e_0 \rangle < 0$ (Q^+ the even sector of Q) and $\varepsilon_0 = 0$ otherwise, and $\varepsilon_\infty = 1$ if $c' \langle \beta \,|\, \xi \rangle < 0$ and $\varepsilon_\infty = 0$ otherwise, c' being the coefficient of s^{2N-1} in P_ξ.

Proof. (a) By Lemma 4, $c = \langle \eta \,|\, \xi \rangle \neq 0$. By Lemmas 2 and 3, D'_η is Q_λ-self-adjoint and annihilates ξ; on $V = \mathbb{R}^I / \mathbb{R}\xi$ its eigenvalues are the $2N$ zeros of P_ξ, simple and non-zero. For a real zero s, $D'_\eta v_s = s\, v_s$, because $\langle \eta \,|\, v_s \rangle = c + sF(s) = c$. The form induced by Q_λ on V is non-degenerate with negative index κ. Lemma 6 gives $\operatorname{sign} Q_\lambda(v_s, v_s) = -\operatorname{sign}\big(P'_\xi(s) P_{\beta\odot\xi}(s)\big)$, and Lemma 7 gives (16).

(b) Take $\varphi_t = \eta + t\beta$ with small $t \neq 0$. On V the eigenvalues of D'_{φ_t} are the $2N$ zeros of $P_t = P_\xi + tP_{\beta\odot\xi}$: the zero $s = 0$ (both polynomials vanish there for every t), $2N - 2$ zeros near the non-zero zeros of P_ξ, and one zero $s_\infty(t) \sim -c'/(t\langle \beta \,|\, \xi \rangle)$. Lemma 6 at a zero of P_t gives the sign $-\operatorname{sign}(P'_t P_{\beta\odot\xi})(s)$, which tends to the sign in (15) as $t \to 0$; at $s_\infty(t)$ the leading terms give the sign of $c' \langle \beta \,|\, \xi \rangle$. For $s = 0$ the eigenvector is the class of $w = D^+\xi - 2\mu e_0$, $\mu = \sum_{k>0} \xi_k / k$: one checks $D'_{\varphi_t} w = \xi$ and $Q_\lambda w \in \mathbb{R}e_0$, so $Q_\lambda(w, w)$ has the sign of $\langle e_0 \,|\, (Q^+ - \lambda)^{-1} e_0 \rangle$. Lemma 7 gives (17). □

Corollary 5.1 (Bezout form)**.** *In Theorem 5(a) let P_ξ and $P_{\beta\odot\xi}$ be coprime, and let B be their Bezout matrix, $\sum_{i,k} B_{ik} x^i y^k = \big(P_\xi(x) P_{\beta\odot\xi}(y) - P_\xi(y) P_{\beta\odot\xi}(x)\big)/(x - y)$. Then B has exactly κ positive and $2N - \kappa$ negative eigenvalues; equivalently, the form induced by $Q - \lambda$ on $\mathbb{R}^I/\mathbb{R}\xi$ is congruent to $-B$.*

Proof. By Hermite's theorem [18] the signature of B is the Cauchy index of $P_{\beta\odot\xi}/P_\xi$, which equals $d - (r - d)$ with r the number of real zeros of P_ξ, and its rank is $2N = r + 2N_c$. Hence the positive index is $N_c + d = \kappa$, and Sylvester's law of inertia gives the congruence. □

Remark 12.1. Since $d(\xi) \geq 0$, Theorem 5 contains Theorem 1; for $\kappa = 0$ it contains Theorem 4. The count can be evaluated without computing any zero, through the inertia of B.

13 Theorem 6: the resolvent lock

Remark 13.1 (Shift invariance). For $c \in \mathbb{R}$, $Q + cI$ is again of CvS type, with the same eigenvectors, the same polynomials P_ξ, $P_{\beta\odot\xi}$ and the same counts $\kappa(\lambda_j)$; only the eigenvalues move by c. The sign of λ_0 is therefore invisible to the zeros of eigenvectors, and is recovered from the resolvent.

Theorem 6 (Resolvent lock)**.** *Let Q be of CvS type, let $\mu \in \mathbb{R}$ be distinct from all eigenvalues, and let $x = (Q-\mu)^{-1}e_0$, $x_0 = \langle e_0 \,|\, (Q-\mu)^{-1}e_0\rangle$. Assume $x_0 \neq 0$, $\langle \eta \,|\, x\rangle \neq 0$, and that the zeros of P_x are simple and lie outside I. Then*

$$N_c(P_x) + d(x) + [x_0 < 0] = \kappa(\mu), \tag{18}$$

where $\kappa(\mu)$ is the number of eigenvalues of Q below μ and $d(x)$ is defined by (15) with ξ replaced by x. In particular, all eigenvalues of Q exceed μ if and only if P_x has only real zeros, $d(x) = 0$ and $x_0 > 0$; for $\mu = 0$ this characterises $\lambda_0 > 0$.

Proof. Put $Q_\mu = Q - \mu$; it is non-degenerate with negative index $\kappa(\mu)$. Since $De_0 = 0$, (4) gives $Q_\mu Dx = DQ_\mu x + (\eta \wedge \beta)x = \eta\langle\beta \,|\, x\rangle - \beta\langle\eta \,|\, x\rangle \in \operatorname{span}\{\eta, \beta\}$, so the argument of Lemma 3 shows that $D'_\eta = D - |Dx\rangle\langle\eta|/\langle\eta \,|\, x\rangle$ is Q_μ-self-adjoint, now on all of $\mathbb{R}^I$. By Lemma 2 its eigenvalues are 0, with eigenvector x of norm $Q_\mu(x,x) = x_0$, and the $2N$ zeros of P_x; since x is even, $P_x(0) = x_0 \prod_{j\neq 0} j \neq 0$. For a real zero s, $v_s = (D-s)^{-1}Dx = x + s\,w_s$ satisfies $D'_\eta v_s = s\,v_s$. Repeating Lemma 6 with $Q_\mu x = e_0$ gives $(D-s)Q_\mu w_s = F\beta - G\eta + e_0$ and two additional contributions: $Q_\mu(x,x) + 2sQ_\mu(x,w_s) = x_0 - 2x_0$ and $s^2\langle(D-s)^{-1}w_s \,|\, e_0\rangle = x_0$. They cancel, so (14) holds unchanged, and Lemma 7 gives (18). □

Remark 13.2. Theorem 6 is the CvS counterpart, with the defect count, of Krein's theorem on polynomials orthogonal with respect to an indefinite weight [15] and of its continuous version by Krein and Langer [4]. Applied to every window, the condition at $\mu = 0$ is equivalent to positivity of the corresponding Weil forms.

14 Certified computations for truncated Weil forms

The Legendre-basis Weil forms of Section 10 (two sectors, $m = 80$ per sector, $F = 400$, 140 digits) are transferred to the Fourier basis U_n, $|n| \leq M$, of the CvS framework through

$$\langle \phi_\ell \,|\, U_n\rangle = (-1)^n\sqrt{2\ell+1}\; i^\ell\, j_\ell(\pi n), \tag{19}$$

with ϕ_ℓ the normalized Legendre functions of the window and j_ℓ spherical Bessel functions. The resulting matrices are of CvS type to within the relative residuals of Table 3; a random perturbation of relative size 10^{-6} of the Legendre matrices raises the residual to $4\cdot10^{-6}$–$7\cdot10^{-6}$.

Certificate. After exact parity symmetrisation, $P_\xi(s) = A(s^2)$ (even ξ) or $sA(s^2)$ (odd ξ), with $\deg A = K$ certified by interval arithmetic. If A shows K sign changes on $(0,\infty)$, evaluated in interval arithmetic, then all zeros of P_ξ are real and simple: the K positive zeros and their mirror images exhaust the degree. The evaluation points are the zeros of $P_{\beta\odot\xi}$ (Theorem 4) or midpoints between computed zeros; the conclusion depends on the signs only. In every cell the certificate detects a zero pair of a certified polynomial that has been moved off the real axis.

Index law. All 237 eigenvectors of Table 4 satisfy Theorem 5, both sectors included; in the odd sector $\varepsilon_0 = 1$ in all 114 cases. Random CvS matrices ($N = 5, 6$): 84/84 even and 72/72 odd eigenvectors, and 66/66 in the self-test of the live run; random γ-symmetric matrices outside the CvS class: 5/84, 3/72 and 10/66. The Bezout form (Corollary 5.1) gives 56/56 for random CvS matrices, 8/56 outside the class, and 24/24 for the Weil truncations at $q = 13$, $M = 10, 12$.

Minuscule block. In every cell all eigenvectors with eigenvalue below 10^{-3} (between 12 and 24 of them) carry the counting certificate: their transforms have only real, simple zeros.

Table 3: Ground-state zero certificates for truncated Weil forms in the Fourier basis (live run, 18 October 2026; $m = 80$ per sector, $F = 400$, 140 digits). Link: CvS residual below 10^{-6} times the spectral gap and λ_0 above the builder floor; otherwise the certificate refers to the computed polynomial.

q	M	CvS residual	certified zeros	link	$\lvert z - \gamma_1 \rvert$
13	10	$2.39 \cdot 10^{-96}$	10/10	yes	$3.7 \cdot 10^{-21}$
13	12	$4.97 \cdot 10^{-84}$	12/12	yes	$1.2 \cdot 10^{-24}$
13	16	$1.22 \cdot 10^{-64}$	16/16	yes	$2.0 \cdot 10^{-30}$
13	20	$5.48 \cdot 10^{-50}$	20/20	yes	$2.3 \cdot 10^{-35}$
29	10	$1.65 \cdot 10^{-96}$	10/10	yes	$2.1 \cdot 10^{-24}$
29	12	$3.60 \cdot 10^{-84}$	12/12	yes	$1.2 \cdot 10^{-29}$
29	16	$7.62 \cdot 10^{-65}$	16/16	yes	$6.9 \cdot 10^{-38}$
29	20	$2.85 \cdot 10^{-50}$	20/20	polynomial	$2.1 \cdot 10^{-45}$
61	10	$5.46 \cdot 10^{-97}$	10/10	yes	$9.0 \cdot 10^{-25}$
61	12	$1.46 \cdot 10^{-84}$	12/12	yes	$1.4 \cdot 10^{-31}$
61	16	$3.19 \cdot 10^{-65}$	16/16	yes	$2.1 \cdot 10^{-42}$
61	20	$1.74 \cdot 10^{-50}$	20/20	polynomial	$7.2 \cdot 10^{-52}$

Table 4: Index law (Theorem 5) for all eigenvectors of the Fourier truncations (live run, 18 October 2026; same settings as Table 3). Signs certified by interval arithmetic; undecided signs: 0. Minuscule block: eigenvalues below 10^{-3}. Capacity: largest $\kappa/2$ for which all positive defect zeros of the even eigenvector lie below γ_1.

q	M	index law	certified real-rooted	minuscule block	capacity
13	10	21/21	16/21	12/12	4
13	12	25/25	18/25	14/14	4
13	16	33/33	20/33	16/16	4
29	10	21/21	20/21	15/15	6
29	12	25/25	22/25	17/17	5
29	16	33/33	26/33	22/22	5
61	10	21/21	20/21	16/16	8
61	12	25/25	24/25	19/19	7
61	16	33/33	27/33	24/24	6

Resolvent lock. Table 5 applies Theorem 6 at $\mu = 0$. In every cell the lock is closed and agrees with the sign of λ_0 obtained independently from the eigenvalues; for the shifted matrices the count returns $k = 1, 2, 4$ exactly (27/27). Random CvS matrices satisfy (18) in 60/60 and 30/30 cases (sandbox and live self-test), random matrices outside the class in 9/60 and 2/30.

15 Observations on the excited states

Measured at $q \in \{13, 29\}$, $M \in \{10, 12\}$ ($m = 60$, $F = 260$, 100 digits), and for the capacity in Table 4:

(i) The zeros of $\hat{\xi}_j$ within 10^{-3} of zeta zeros coincide with those of the ground state throughout the minuscule block.

(ii) For even ξ_j the $\kappa/2$ positive defect zeros first fill the zero-free interval $(0, \gamma_1)$, $\gamma_1 = 14.1347\ldots$, as an approximately half-integer ladder $z_i \approx (i - \frac{1}{2})\Delta$; afterwards they enter the gaps between consecutive zeta zeros.

(iii) The capacity of $(0, \gamma_1)$ decreases with M: for $M = 10, 12, 16, 20$ it is $4, 4, 4, 4$ at $q = 13$,

Table 5: Resolvent lock (Theorem 6) at $\mu = 0$ for the Fourier truncations (live run, 18 October 2026; same settings as Table 3). Lock: P_x certified real-rooted, $d(x) = 0$ with interval-certified signs, $x_0 > 0$. Shifted matrices $Q - cI$ with $k = 1, 2, 4$ negative eigenvalues: in every cell $\kappa(0) = k$ equals $N_c + d + [x_0 < 0]$, namely $1 = 0 + 0 + 1$, $2 = 1 + 0 + 1$, $4 = 1 + 2 + 1$.

q	M	λ_0	x_0	$x_0\lambda_0$	lock	shifted ($k = 1, 2, 4$)
13	10	$2.83 \cdot 10^{-26}$	$1.17 \cdot 10^{25}$	0.332	closed	3/3
13	12	$2.14 \cdot 10^{-29}$	$1.5 \cdot 10^{28}$	0.322	closed	3/3
13	16	$8.57 \cdot 10^{-35}$	$3.62 \cdot 10^{33}$	0.31	closed	3/3
29	10	$8.92 \cdot 10^{-32}$	$3.34 \cdot 10^{30}$	0.299	closed	3/3
29	12	$3.54 \cdot 10^{-36}$	$8.03 \cdot 10^{34}$	0.285	closed	3/3
29	16	$1.62 \cdot 10^{-43}$	$1.65 \cdot 10^{42}$	0.267	closed	3/3
61	10	$2.61 \cdot 10^{-35}$	$1.08 \cdot 10^{34}$	0.283	closed	3/3
61	12	$2.0 \cdot 10^{-40}$	$1.33 \cdot 10^{39}$	0.267	closed	3/3
61	16	$1.29 \cdot 10^{-49}$	$1.91 \cdot 10^{48}$	0.246	closed	3/3

$6, 5, 5, 4$ at $q = 29$ and $8, 7, 6, 5$ at $q = 61$ (the $M = 20$ values measured with $m = 100$). Its large-M behaviour is the subject of the next note.

(iv) In all cells of Table 5, $x_0\lambda_0$ lies between 0.246 and 0.332, decreasing in q and M. This matches the dominance of the ground state in $x = Q^{-1}e_0$: $x \approx (\xi_0)_0\, \xi_0/\lambda_0$, so the sign part of the lock follows the sign of λ_0 and its zero part follows Theorem 4; the zero part becomes decisive when several eigenvalues cross 0 together, as in the shifted matrices with $k \geq 2$.

Heuristically, the Weil form behaves like a sum of $|\hat{f}(\gamma)|^2$ over the zeta zeros in the band. Eigenvectors with minuscule eigenvalue then vanish near every such zero, and the κ additional zeros required by Theorem 5 are placed in the intervals free of zeta zeros, beginning with $(0, \gamma_1)$. Connes and Consani [20] describe these eigenvectors through prolate spheroidal wave functions; (i)–(iii) record their zeros.

Author's contributions

The research programme behind this note—studying the sinh face of the pole term alongside the cosh face, and requiring every real-rootedness statement for truncated Weil forms to be established in both parity sectors—was formulated by the author, and the two-sector measurement pipeline used in Section 10 was built under his direction. Theorems 1–3 answer the question posed by this programme: whether the Connes–van Suijlekom real-zeros mechanism persists across the change of sector and beyond the ground state. The requirement that every eigenvector, and not only the ground state, be accounted for by an exact count; the use of pre-registered predictions with negative controls for every computation; and the certification standard of Section 14 (interval arithmetic with a counting argument, plus a separate gate for the link to the Weil truncation) were set by the author. The programme of reaching $\lambda_0 > 0$ through a zero-free lock at $\kappa = 0$ was proposed by the author; Theorem 6 realises it through the resolvent.

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Data availability

Verification scripts and high-precision data are available from the author on request; the certification tools are `icice_sertifika_v1.py` (md5 `1b7dea17…`) , `icice_sertifika_v1_1.py` (md5 `3d720f0b…`) and `icice_sertifika_v1_2.py` (md5 `fe2ddcb2…`), with their logs of 18 October 2026.

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